Tara Sawrikar

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EDUCATION

Harvey Mudd College, Claremont, CA
B.S. Computer Science and Mathematics, Concentration in Art | Dean's List — Expected May 2029
Relevant Coursework: Linear Algebra, Discrete Mathematics, Differential Equations, Mechanics and Wave Motion

TECHNICAL SKILLS

Languages: C++ (C++17), Python, C#, GLSL, TypeScript, JavaScript, Java
Graphics & DCC: OpenUSD, OpenGL, WebGL, physically based rendering (PBR), Autodesk Maya, Blender, Unity
APIs & Tools: Maya Python API, Unity Editor API, React, Vitest, WebAssembly, Dear ImGui, NumPy, pandas, CMake, Git

INTERNSHIP EXPERIENCE

Software Engineering Intern | **QSimulate**, Boston, MA — May 2026 – Aug 2026
- Built production React/TypeScript features for interactive 3D visualization of quantum-simulation results, rendering molecular structures and orbitals with the NGL WebGL viewer, WebAssembly and JSON scientific data.
- Led implementation of Vitest screenshot-testing infrastructure, adding visual regression coverage across the full platform; the framework was merged into production and adopted by the team.
- Wrote component and integration tests for 9+ scientific workflows.

Engineering Intern | **Clean Properties Engineering, Inc.**, Sudbury, MA — Jun 2025 – Aug 2025
- Produced site-assessment reports and analyzed environmental data supporting wastewater treatment projects.

Research Intern | **Weston Observatory**, Boston College — Jun 2023 – Aug 2024
- Maintained earthquake-monitoring systems and restored laboratory infrastructure supporting seismic data collection.

SELECTED PROJECTS EXPERIENCE

OpenUSD Physically Based Renderer | *C++, OpenUSD, Blender, OpenGL, GLSL* | — 2026
- Built a real-time C++17/OpenGL 4.1 renderer that loads Blender-authored OpenUSD scenes via the official OpenUSD SDK, rendering a 17-mesh, 7,904-triangle scene with no manual conversion.
- Wrote a USD scene loader that traverses the stage, triangulates meshes, and rebuilds transform hierarchies, normals, UVs, and UsdPreviewSurface materials into a scene graph with automatic camera framing.
- Implemented a Cook-Torrance PBR shader in GLSL (GGX, Smith geometry, Schlick Fresnel) for metallic/roughness materials with ACES tone mapping and gamma correction, plus a Dear ImGui diagnostics overlay.

Production Asset Publisher for Autodesk Maya | *Python, Maya Python API, OpenUSD* | 2026
- Built an artist-facing publishing tool that blocks publishing until 6 scene validators pass (naming, root selection, frozen transforms, construction history, duplicate names, empty groups).
- Automated one-click publishing into versioned folders (v001, v002, ...) with Maya ASCII and OpenUSD export, viewport thumbnails, and JSON metadata, using a modular, JSON-configured architecture.

Unity Asset Validation Framework | *C#, Unity Editor API* | 2026
- Built a Unity 6 Editor tool with modular validators that scans a project's textures, materials, and prefabs for missing scripts, missing textures, invalid names, and duplicate asset names.
- Added severity filtering, search, one-click asset selection, Quick Fix actions for naming issues, a project health dashboard, and CSV reports with suggested fixes.

RESEARCH EXPERIENCE

Student Researcher | Creative Computing Studio, Harvey Mudd College — Sept 2026 – Present
- Co-developing a browser-based studio for Bantam Tools NextDraw and AxiDraw pen plotters in Python (FastAPI) and JavaScript that previews scripts and SVGs, estimates plot time, and runs live plots.
- Built an image-to-plot converter with CMYK color separation and 10 drawing styles (hatching, stippling, flow fields, contour lines), plus single-line text and a phone drawing pad synced over Wi-Fi.

Student Researcher | FICUS Lab, Harvey Mudd College — Jan 2026 – Present
- Implementing ecosystem-specific ML models in Python (scikit-learn, pandas) on top of a general carbon-flux pipeline, comparing them across 206 FLUXNET sites with leave-one-site-out validation.
- Analyzed eddy-covariance CO2 flux and rainfall data in Python to study rainfall-triggered respiration pulses; won the Outstanding Student Poster Award.

LEADERSHIP

Activities: Leadership Team, Gender Equity in Math | Treasurer, Harvey Mudd Racing | Leadership Team, Adaptive Design at Mudd | Leadership Team, Harvey Mudd Science Day | Grader + Tutor, CS5
Awards: Bonnie Scholarship Award (2025) | The Village Bank Scholarship (2025) | Girl Scout Gold and Silver Award (2024)